CSA 1718 – Artificial Intelligence

Name : P. Yaswanth

Reg No : 192524167

EXPERIMENT : Finding the Number of Vowels Using Prolog

Aim

To write and execute a Prolog program to find the **number of vowels present in a given word or list of characters**.

Objective

- To identify vowels using Prolog facts.
- To count vowels recursively.
- To understand recursion and list processing in Prolog.

Theory

A vowel is one of the characters **a, e, i, o, u**. In Prolog, we can define these vowels as facts. The input word can be represented as a list of characters.

For example:

[h,e,l,l,o]

contains two vowels: e and o.

Algorithm

1. Start the program.
2. Define the vowels a, e, i, o, u.
3. Take a list of characters as input.
4. Check the first character.
5. If it is a vowel, increase the count by 1.
6. If it is not a vowel, keep the count unchanged.
7. Recursively process the remaining characters.
8. When the list becomes empty, return the final count.
9. Stop.

Prolog Program

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```
% Vowel definition

vowel(a).
vowel(e).
vowel(i).
vowel(o).
vowel(u).

% Base case

count_vowels([], 0).

% If the first character is a vowel

count_vowels([H|T], Count) :-
    vowel(H),
    count_vowels(T, C),
    Count is C + 1.

% If the first character is not a vowel

count_vowels([H|T], Count) :-
    \+ vowel(H),
    count_vowels(T, Count).
```

Queries and Output

Query 1:

?- count_vowels([h,e,l,l,o], N).

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Output:

N = 2.

Query 2:

?- count_vowels([a,p,p,l,e], N).

Output:

N = 2.

Query 3:

?- count_vowels([c,o,m,p,u,t,e,r], N).

Output:

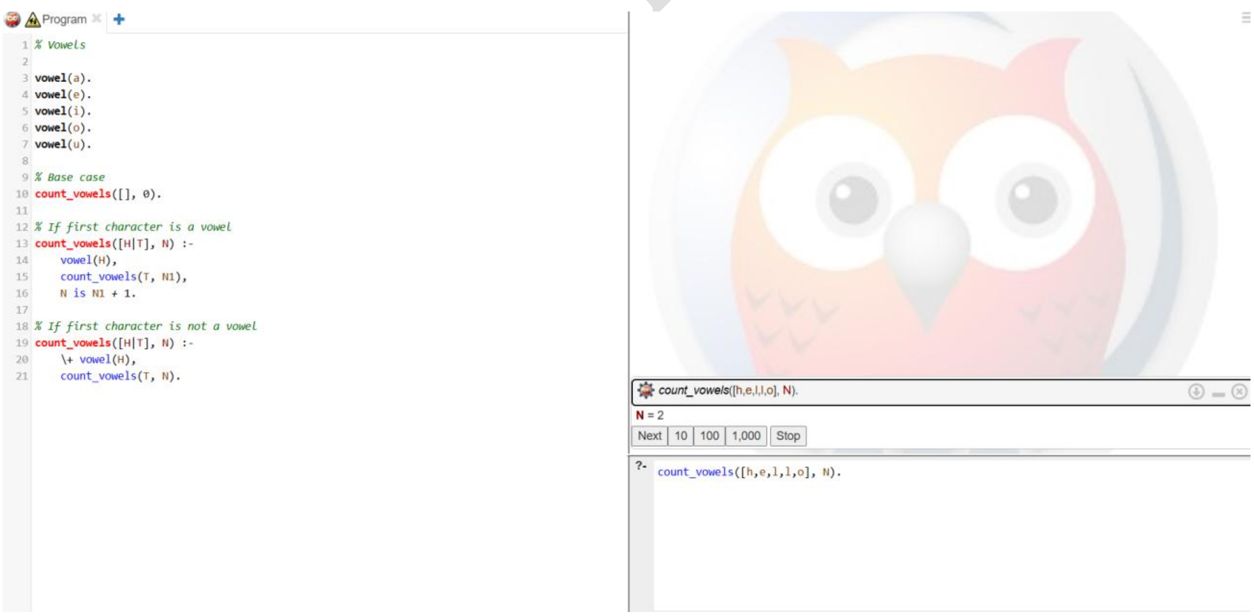
N = 3.

Query 4:

?- count_vowels([p,r,o,g,r,a,m], N).

Output:

N = 2.



Explanation

- vowel(a)., vowel(e)., etc. define the vowels.
- count_vowels([], 0) is the **base case**. When there are no characters left, the count is zero.

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- [H|T] divides the list into:
    - H → first character
    - T → remaining characters
- If H is a vowel, Count is increased by 1.
- If H is not a vowel, the count remains unchanged.
- Recursion continues until the complete list is processed.

Applications

Vowel counting can be used in:

- Text processing
- Natural Language Processing
- String analysis
- Language-learning applications
- Basic AI text-processing systems

Result

Thus, the **Prolog program for finding the number of vowels in a given list of characters** was written and executed successfully.

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